

Md Shamim Hussain, Ph.D.

ML Research Scientist

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Website | GitHub | LinkedIn | Google Scholar

Professional Summary

Machine learning research scientist developing deep neural architectures for graph-structured data, particularly molecular graphs. Research spans evolving structural representations, higher-order geometric interactions, 2D-to-3D molecular learning, and efficient global attention, with first-author publications at ICML and KDD. Experienced in multi-stage pretraining, fine-tuning, transfer learning, and distributed GPU training.

Core Expertise

Deep Neural Architecture Design | Graph Transformers | Graph Neural Networks | Geometric Deep Learning | Molecular Geometry and Property Prediction | Distributed GPU Training and Fine-Tuning

Experience

Synlico Inc.

Machine Learning Scientist

Oct 2025 - Present

South San Francisco, CA

- Research deep reinforcement learning-based causal discovery methods to infer gene regulatory networks as directed causal graphs from high-dimensional single-cell genomics data.
- Develop GPU-accelerated training and evaluation workflows for causal structure learning over gene networks.
- Collaborate across machine learning and bioinformatics to interpret inferred causal graphs.

Rensselaer-IBM AI Research Collaboration

AI Horizons Scholar; IBM Resident Scholar in 2022 and 2023

Aug 2020 - Jul 2025

Troy, NY / Yorktown Heights, NY

- Developed the **Triplet Graph Transformer (TGT)**, extending EGT with third-order triplet interactions among groups of three atoms and a 2D-to-3D framework that predicts interatomic distances from 2D molecular graphs for downstream property prediction; produced the first-ranked single-model submission on the PCQM4Mv2 molecular-property prediction benchmark (a position retained for more than two years) and the leading direct-method result on the OpenCatalyst 2020 - IS2RE task.
- Designed a multi-stage training pipeline comprising distance prediction, noisy-coordinate pretraining, and fine-tuning; trained molecular transformer configurations up to 102M parameters.
- Scaled experiments across multiple GPUs and nodes using PyTorch, `torchrun`, and Slurm on the AiMOS supercomputer; released code, model weights, datasets, and inference tools.
- Developed the **Edge-augmented Graph Transformer (EGT)**, an early pure graph transformer architecture introducing evolving structural edge channels for joint node and pair representations across graph-level prediction, edge classification, and link prediction.
- Proposed **Stochastically Subsampled Self-Attention (SSA)** and formulated the Information Pathways Hypothesis, reducing transformer self-attention memory and computation by 4-8x while improving generalization across graph, vision, and language tasks.

Bangladesh University of Engineering and Technology

Research Engineer, DSP Research Lab

Aug 2017 - Jul 2019

Dhaka, Bangladesh

- Designed **SwishNet**, a lightweight 1D CNN using knowledge distillation for CPU-efficient audio classification, and developed a bimodal-ultrasound breast-lesion classifier using transfer and multi-task learning.

Selected Research Contributions and Publications

Triplet Interaction Improves Graph Transformers: Accurate Molecular Graph Learning with Triplet Graph Transformers.

ICML 2024, PMLR 235:20768-20792. Paper | Code | Models

First-author work introducing third-order interactions and a geometry-aware, multi-stage molecular learning framework; released reproducible training and inference code, pretrained weights, and processed PCQM4Mv2 data.

The Information Pathways Hypothesis: Transformers are Dynamic Self-Ensembles.

KDD 2023, pp. 810-821. DOI: 10.1145/3580305.3599520 | arXiv

First-author work introducing stochastic attention subsampling and the Information Pathways Hypothesis for efficient transformer training and improved generalization.

Global Self-Attention as a Replacement for Graph Convolution.

KDD 2022, pp. 655-665. DOI: 10.1145/3534678.3539296 | Code

First-author work introducing evolving structural edge channels and demonstrating that global self-attention can replace localized convolution for general-purpose graph learning.

Selected Open-Source Engineering

Triplet Graph Transformer - Author and maintainer of the official PyTorch implementation, including 102M-parameter configurations, multi-node training, staged checkpointing, inference workflows, pretrained weights, and processed molecular data. (GitHub)

Edge-augmented Graph Transformer - Author and maintainer of the PyTorch and TensorFlow implementations, including distributed training workflows, pretrained molecular checkpoints, and transfer-learning configurations for OGB benchmarks. (GitHub)

Technical Skills

Research: Graph transformers, graph neural networks, geometric deep learning, molecular property prediction, transformer architectures, reinforcement learning

Machine Learning: PyTorch, TensorFlow, RDKit, scikit-learn, Hugging Face, OpenCV, NumPy, pandas

Systems: CUDA, PyTorch distributed training, torchrun, Slurm, MPI, Docker, Git, Linux, Bash

Programming and Data: Python, C, C++, MATLAB, SQL, Numba, PyArrow

Education

Rensselaer Polytechnic Institute - Ph.D. in Computer Science (2025)

Rensselaer Polytechnic Institute - M.S. in Computer Science (2021)

Bangladesh University of Engineering and Technology - M.S. in Electrical and Electronic Engineering (2019)

Bangladesh University of Engineering and Technology - B.S. in Electrical and Electronic Engineering (2017)

Professional Service and Recognition

Reviewer for *Data Mining and Knowledge Discovery* (2025), *IEEE Transactions on Neural Networks and Learning Systems* (2023), and *IEEE Transactions on Big Data* (2023). IBM AI Horizons Scholarship recipient (2020).